

Final Exam Checklist

Before the Exam

- **Types of questions:** Multiple choice, short answer questions (about the understanding key concepts and algorithms). The exam will NOT ask for writing codes/pseudocodes.
- **Practice:** Quizzes are simulations of multiple-choice questions in the exam. Assignments are good examples for preparing for the short answer questions. Revisit the definitions of key concepts in the lecture notes to prepare for some short answer questions related to concepts and definitions.
- **Plan your time:** Create a short study schedule up to the exam day.

Content Review

- **Search (~40%):**
 - Systematic search:
 - * State-based representation: Know how to formulate a practical problem as a state-based search problem: **state representation, actions, transitions, cost**.
 - * Uninformed Search:
 - Understand the basic idea of DFS, BFS, UCS, ID algorithms;
 - Know the time and space complexity, completeness (can find a goal or not);
 - Comparison among different algorithms.
 - * Informed Search:
 - Understand and apply **A* search algorithm**, understand the basic idea of DF Branch-and-Bound algorithm;
 - Understand what is **admissibility** of a heuristic and know how to determine a heuristic function is admissible or not.
 - Local search:
 - * Feature-based representation:
 - Know how to formulate a practical problem as a search problem in **feature space**.
 - Know how to define the **neighborhood** and the **valuation/loss function** of a candidate solution.
 - * Local search:
 - Understand and apply **Iterative best improvement (Greedy Descent)**;
 - Understand the limitations of local search and ideas to improve local search.
 - * Stochastic local search:

- Understand the basic idea of simple stochastic approaches RII and PII;
 - Understand the basic idea of Simulated annealing, and know how the temperature T controls the tolerance of worsening steps;
 - Understand and apply **Gradient descent** algorithms for local search and know how the learning rate affects the search process.
- Adversarial search:
 - * Know what a game tree is.
 - * Understand and apply **minimax** and **expectimax** algorithm on game trees;
 - * Understand and apply **alpha-beta pruning** technique on a minimax game tree.
 - * Understand different adversary settings: optimal/random agent v.s. optimal/random opponents.
- Genetic algorithm:
 - * Know how to formulate a solution as a string representation;
 - * Familiar with its key operations including selection, crossover and mutation.
- **Knowledge & Logics (~30%):**
 - Knowledge representation:
 - * Propositional logic:
 - Understand the key concepts, including atom propositions, connectives, interpretations, literals, clauses, logical implications, model of a knowledge base;
 - Familiar with the **truth table of some basic compound propositions**, such as $\neg A, A \wedge B, A \vee B, A \rightarrow B$, especially $A \rightarrow B$ is true with any value of B if A is false.
 - * First-order logic:
 - Understand the key concepts of FOL, including variables, constants, relations, and functions;
 - Understand **universal** and **existential quantifiers**. Familiar with the **translations between English and FOL sentences**;
 - Know how to formulate an inference problem with FOL, i.e., definition of relations, variables, sentences, etc.
 - Logical inference:
 - * Propositional logic inference:
 - Understand what a model of KB is. Understand the logical inference problem, i.e., entailment by knowledge base & observation;
 - Familiar with the definition of **definite clause** and **Horn clause**, and their corresponding inference problem;
 - Understand and apply **SLD resolution algorithm**, and compare with the forward chaining algorithm.
 - * FOL inference:
 - Understand the key operations to handle variables and quantifiers **substitution** and **unification**;
 - Understand and apply **backward chaining (resolution)**.

- **Reasoning and Decision-making with Uncertainty (~15%):**

- Familiar with Bayes rule, conditional independence, and **probabilistic inference** on a given problem with CPTs.
- Understand the definition of **Bayesian networks** and **local Markov property** (for determining independence).
- Understand and apply **variable elimination** algorithm to solve simple inference problems in Bayesian networks.
- Understand the basic definition of a decision network and apply the **variable elimination** algorithm on a decision network.
- Understand the definition of an MDP, possible runs of a defined MDP and the γ -discounted expected utility. Understand the basic idea of policy iteration and value iteration algorithms and their difference.

- **Neural Networks (~15%):**

- Understand the definition of an **artificial neuron** and the necessity of non-linear activations.
- Familiar with the key properties of typical **activation functions** (sigmoid, ReLU, LeakyReLU and tanh).
- Understand the **backpropagation** algorithm (forward and backward) and apply it on simple computation graph.
- Compare batch/mini-batch/SGD methods.
- Understand the basic motivation of CNNs and the relations between filter sizes, image sizes, and padding sizes. Understand the motivation of RNNs, attentions and Transformers, and know their capability.

Exam Strategy

- Read instructions carefully.
- Quickly estimate how much time to spend on each section. Start with easier questions to build confidence and secure quick points.
- Write clear steps where partial credit may be given.
- Reserve a few minutes at the end to re-check answers.